

Q4: Discovering Candidate Drug Targets in Melanoma

DepMap dependency screening, SOX10 validation, and druggable proxy-target prioritisation

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Research question

Which genes are selectively required by melanoma cells, and which pharmacologically tractable genes could provide an indirect therapeutic entry point into the SOX10-dependent melanoma state?

1. Analysis overview

The analysis used DepMap 24Q2 CRISPR and expression data as a funnel rather than treating every essential gene as a drug target. Each step removed a specific source of false-positive prioritisation.

1 Define melanoma models	2 Find selective dependencies	3 Validate SOX10	4 Find SOX10 co-dependencies	5 Prioritise druggable proxies
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2. Step-by-step decisions

Decision	What was done	Why it was done
Define melanoma models	Selected models using OncoTree primary disease/subtype = melanoma.	Using the broad "Skin" lineage could include non-melanoma skin cancers and distort selectivity.
Integrate three DepMap layers	Used Chronos gene effect, dependency probability, and RNA expression.	A plausible target should be essential, consistently dependent, and actually expressed in melanoma cells.
Remove common essentials	Excluded DepMap inferred common-essential genes.	Ribosomal, proteasomal, and housekeeping genes may be strongly essential but usually lack a safe tumour-specific therapeutic window.
Measure melanoma selectivity	Compared mean Chronos effects in melanoma versus all other cancer models.	A melanoma target should be more essential in melanoma, not simply essential everywhere.
Test statistical support	Applied Welch tests, Benjamini-Hochberg FDR correction, Hedges g, and dependency-prevalence comparisons.	Fixed cut-offs alone do not quantify uncertainty or effect size.
Rank candidates	Required dependency, selectivity, expression, prevalence, dependency probability, FDR, and non-common-essential status.	The combined filter prevents one extreme metric from dominating the shortlist.
Validate SOX10 robustness	Used 1,000 bootstrap samples, leave-one-model-out analysis, threshold sensitivity, and expression-dependency correlation.	This tests whether SOX10 depends on arbitrary cut-offs, one outlier model, or unstable sampling.
Find proxy targets	Correlated each gene's CRISPR dependency profile with SOX10 across DepMap, then retained melanoma-selective, expressed, non-common-essential genes.	SOX10 is a transcription factor; a co-dependent enzyme or regulator may offer a more tractable route to exploit the same cell state.
Annotate druggability	Separated established controls, mechanistic regulators, poorly druggable transcription factors, exploratory genes, and prioritised tractable candidates.	Co-dependency nominates biology; it does not by itself prove direct regulation or druggability.

3. Main finding: SOX10 is a robust melanoma-selective dependency

After excluding common essentials and applying dependency, selectivity, expression, prevalence and statistical filters, seven candidate dependencies remained. SOX10 ranked first and was taken forward for focused validation.

Metric	Result	Metric	Result
Candidate rank	1	Mean Chronos effect in melanoma	-1.369
Mean effect in other cancers	-0.185	Melanoma selectivity	-1.185
Melanoma models dependent	84.0%	Other-cancer models dependent	2.8%
FDR	1.16×10^{-20}	Hedges g	-4.49
Mean dependency probability	0.851	Melanoma expression prevalence	88.9%

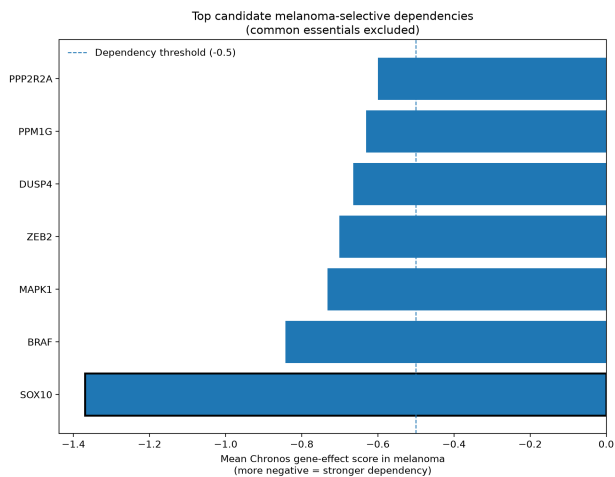


Figure 1. Top filtered melanoma dependencies.

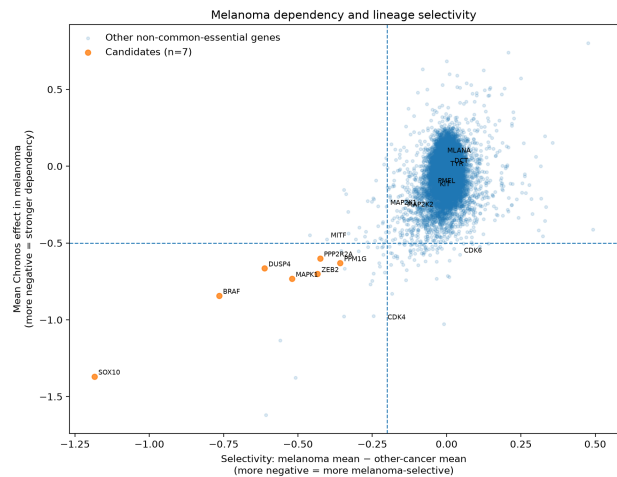


Figure 2. Dependency versus melanoma selectivity.

Decision from Figures 1–2

SOX10 showed the strongest mean melanoma dependency and occupied an isolated lower-left position, combining strong essentiality with high melanoma selectivity. Recovery of BRAF and MAPK1 supported the validity of the screening approach, but these were treated as established positive controls rather than new targets. SOX10 was therefore selected as the primary discovery for robustness testing and proxy-target analysis.

Expression as a potential dependency biomarker



Figure 3. SOX10 expression versus dependency across melanoma cell lines.

Decision from this result

Higher SOX10 expression was associated with a more negative Chronos score (Spearman $\rho = -0.338$, $p = 0.0032$). The relationship is heterogeneous and partly state-like rather than perfectly linear. SOX10 expression was therefore interpreted as a potential biomarker of dependency, not as a universal predictor for every melanoma model.

Dependency prevalence and heterogeneity

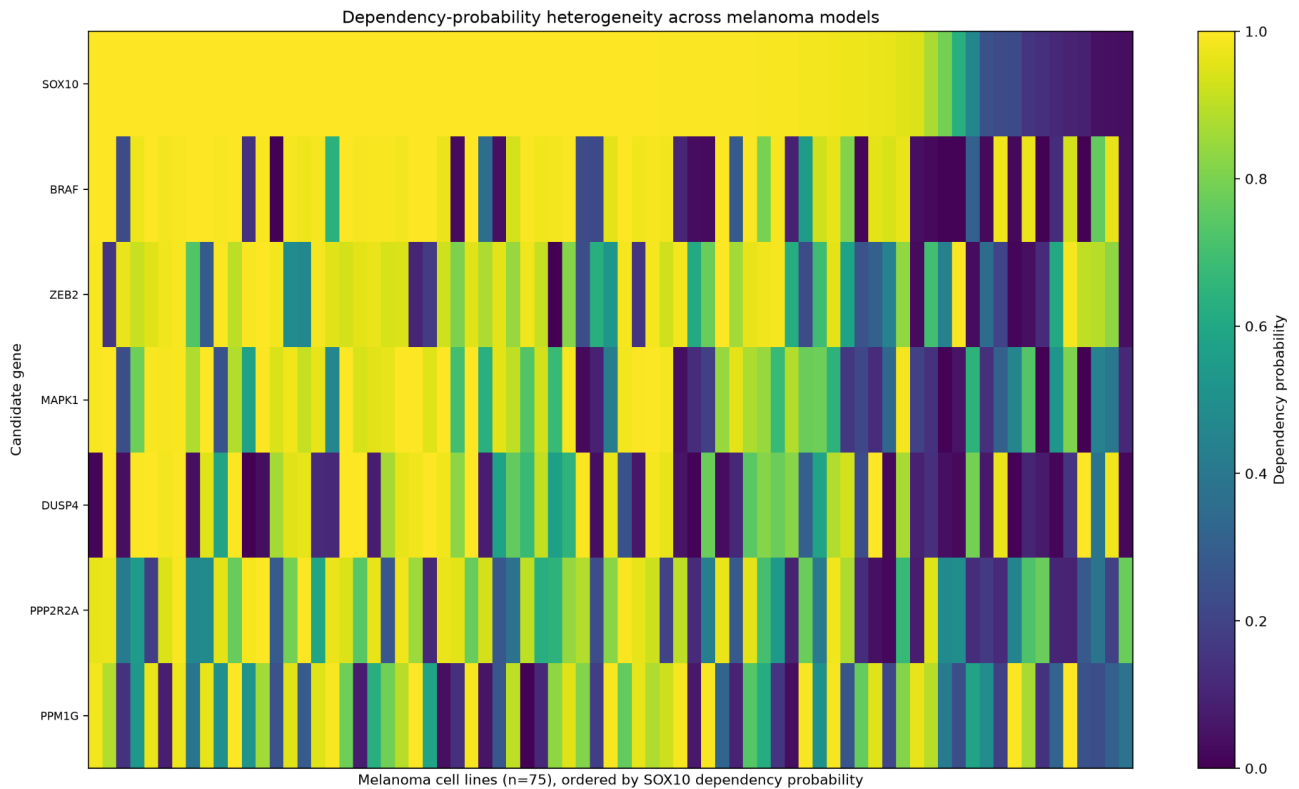


Figure 4. Dependency probability across melanoma models ordered by SOX10 dependency.

Decision from this result

SOX10 dependency probability was high in most melanoma models, consistent with the 84% dependency prevalence, but a smaller SOX10-low or SOX10-independent group remained. This supported SOX10 as a broad lineage vulnerability while motivating future stratification and the search for alternative druggable entry points into the same cell state.

Robustness to sampling and threshold choices

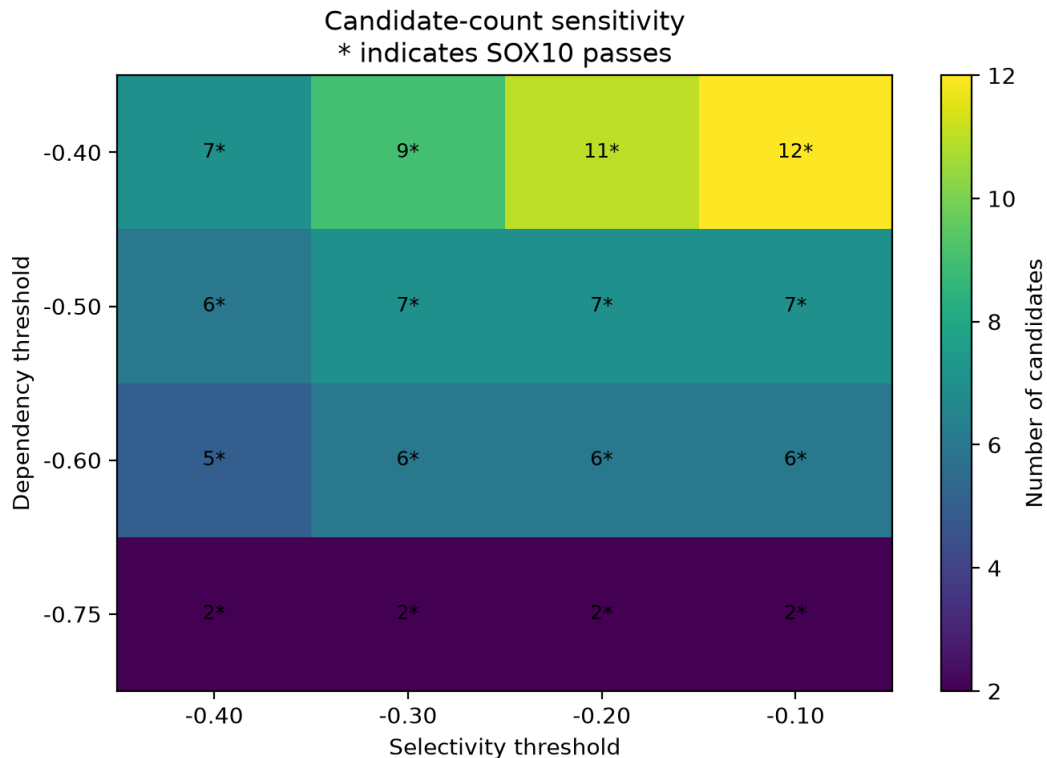


Figure 5. Candidate-count sensitivity across dependency and selectivity cut-offs.

Decision from this result

SOX10 passed every tested threshold combination, including the strictest cut-offs. The conclusion was therefore not dependent on a convenient choice of -0.5 for dependency or -0.2 for selectivity.

Robustness checks

- Bootstrap 95% CI for the melanoma mean effect: -1.525 to -1.222.
- Bootstrap 95% CI for selectivity: -1.339 to -1.036.
- Leave-one-model-out selectivity remained between -1.203 and -1.167.
- SOX10 passed every tested dependency/selectivity threshold combination.

Functional annotation of SOX10

SOX10 was submitted as a single gene to Enrichr and ToppFun to assess biological plausibility. The outputs linked SOX10 to neural-crest differentiation, melanocyte differentiation, pigmentation and MITF-associated transcriptional programmes. Because only one gene was entered, this was treated as cross-database functional annotation rather than conventional gene-set enrichment. The result supported melanoma-lineage relevance but did not independently validate pharmacological tractability.

4. SOX10 proxy-target analysis

SOX10 is a transcription factor and may be difficult to inhibit directly. To identify more tractable therapeutic entry points, each gene's CRISPR Chronos profile was correlated with SOX10 dependency across 1,150 DepMap models. Genes were retained when they also showed melanoma dependency, melanoma selectivity, melanoma expression, significant correlation and exclusion from common essentials. Eighteen genes passed the screening rules.

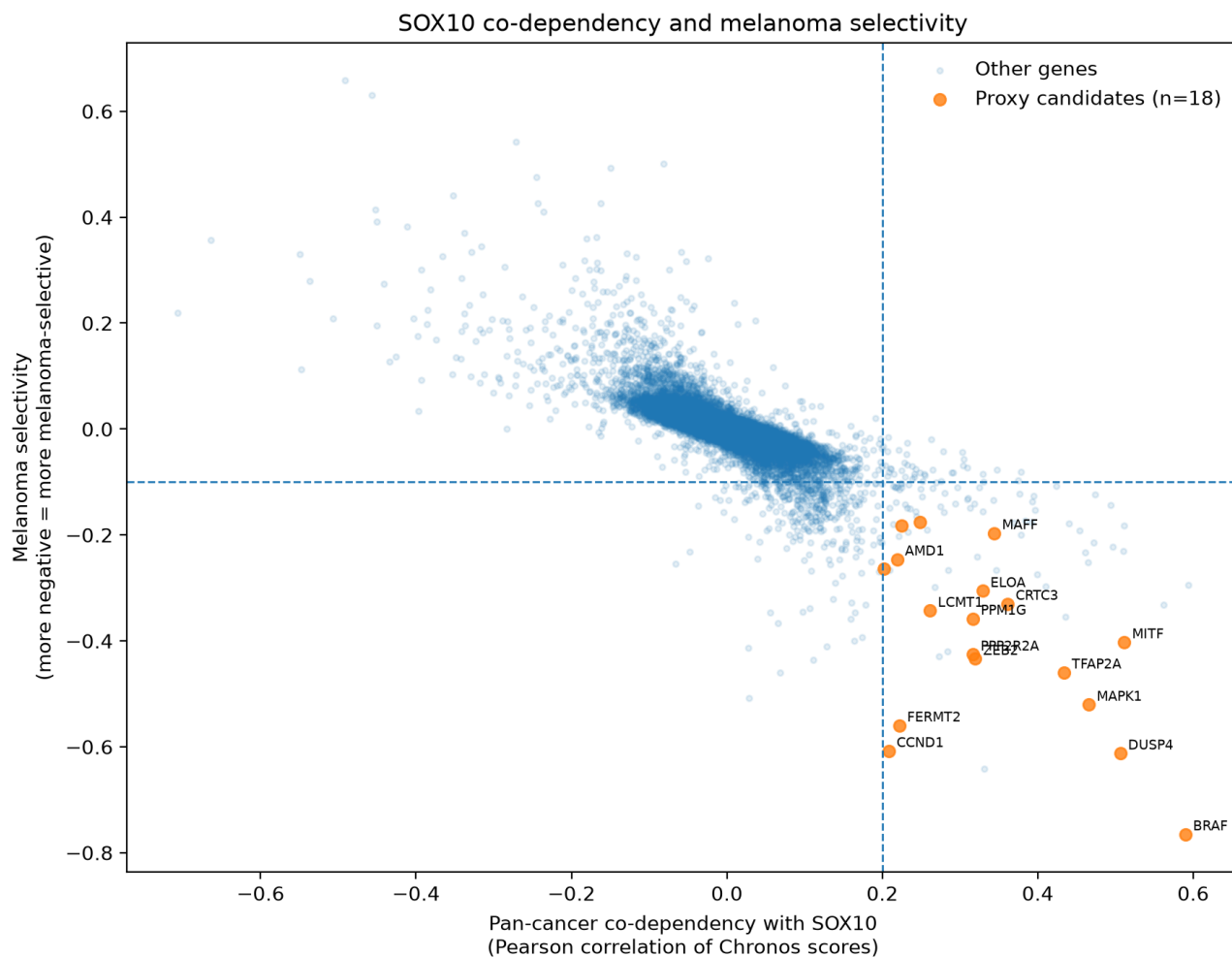


Figure 6. SOX10 co-dependency and melanoma selectivity for the 18 retained proxy candidates.

Decision from this result

The preferred region combines higher positive SOX10 co-dependency with more negative melanoma selectivity. BRAF and MAPK1 ranked strongly and served as positive controls. The next decision was not to accept the numerical ranking alone, but to classify candidates by novelty, target class, known melanoma biology, directionality and pharmacological tractability.

Annotation and deadline-focused prioritisation

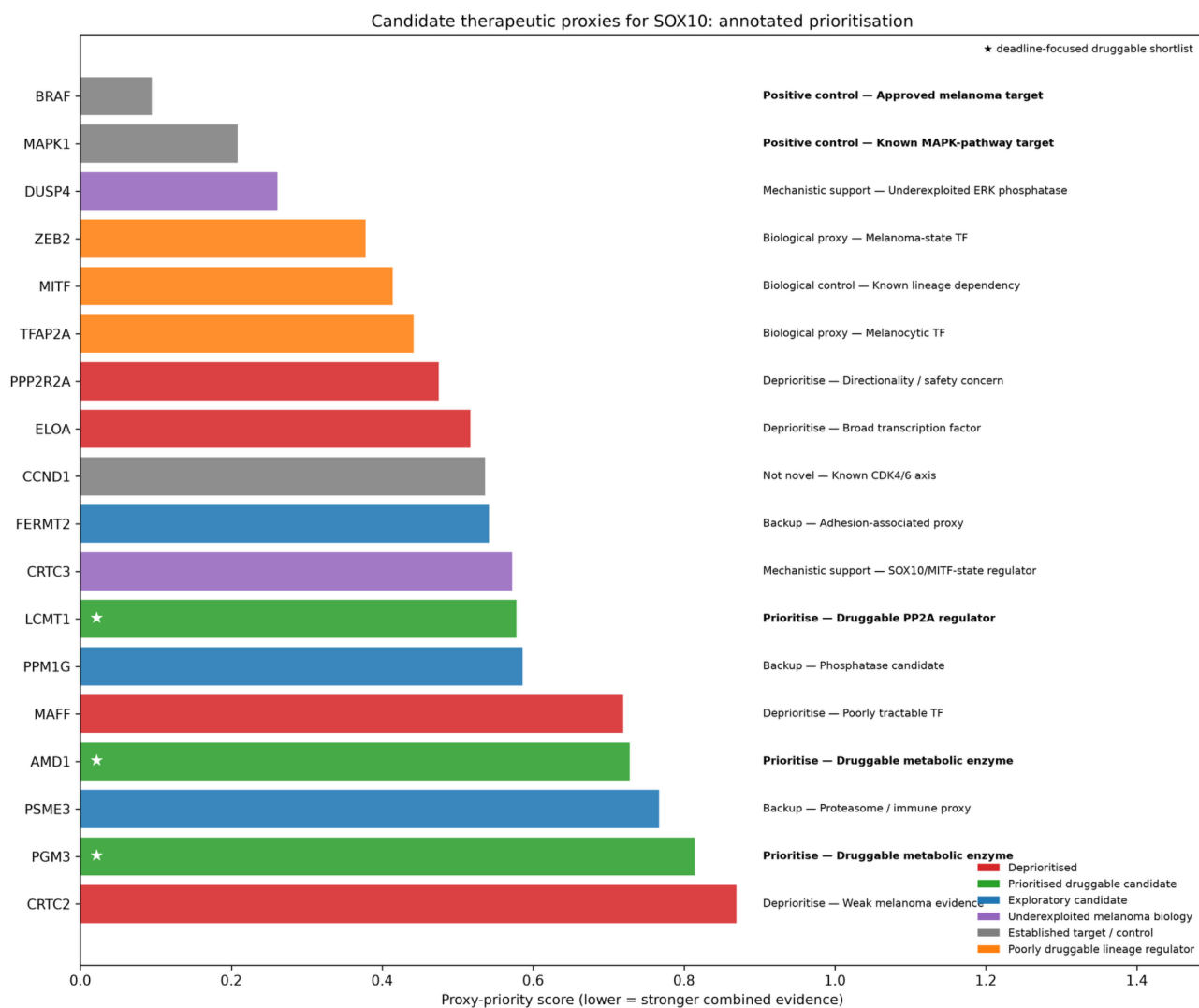


Figure 7. Annotated prioritisation of candidate therapeutic proxies.

Decision from this result

The proxy-priority score summarised computational evidence, where lower values indicate stronger combined support. The green-star shortlist was selected only after a second translational filter. AMD1, PGM3 and LCMT1 were retained because they are enzymatic or regulatory targets with existing inhibitor or preclinical tractability evidence and are less established as melanoma treatment targets. Their selection therefore reflects combined computational evidence, novelty and druggability, not simply the three lowest numerical scores.

Category	Genes	Decision rationale
Established controls	BRAF, MAPK1, CCND1	Recovered known MAPK/cell-cycle biology; validates the method but is not novel.
Mechanistic/underexploited	DUSP4, CRTC3	Strong biological links to the melanoma state, but not the most mature direct drug targets.
Poor direct druggability	MITE, ZEB2, TFAP2A	Biologically relevant transcriptional regulators, but difficult to inhibit selectively.
Prioritised tractable proxies	AMD1, PGM3, LCMT1	Enzymatic/regulatory targets with existing inhibitor or preclinical tractability evidence.
Exploratory	FERMT2, PPM1G, PSME3	Potentially interesting, but weaker mechanistic or pharmacological support.
Deprioritised	PPP2R2A, ELOA, MAFF, CRTC2	Directionality, safety, broad cellular function, or weak melanoma/druggability evidence.

Final focused shortlist

Candidate	Role in shortlist	Why retained
AMD1	Best translational candidate	Metabolic enzyme; a direct inhibitor exists; the melanoma application remains to be tested.
PGM3	Best novel preclinical candidate	Metabolic enzyme with preclinical inhibition evidence in other cancers and little established melanoma targeting.
LCMT1	Early tractable backup	Enzymatic PP2A regulator with early melanoma-relevant pharmacology, but therapeutic-window concerns require follow-up.

5. What can and cannot be claimed

Supported by this analysis	Not yet demonstrated
SOX10 is a strong, selective and reproducible melanoma genetic dependency.	That a proxy gene directly regulates or physically interacts with SOX10.
SOX10 dependency is present across most melanoma models and relates to SOX10 expression.	That inhibiting the candidate will selectively kill patient tumours.
The method recovers known melanoma biology such as BRAF and MAPK1.	That the candidate is clinically safe or superior to current treatments.
AMD1, PGM3 and LCMT1 are candidate tractable proxies associated with the SOX10-dependent state.	That a target is completely novel without a full literature and database audit.

6. Reproducibility and file map

Main scripts: melanoma_depmap_v2.py, sox10_proxy_targets.py, and sox10_proxy_plot_clean.py.

Key outputs: outputs_v2/results/sox10_summary.txt and outputs_proxy/results/sox10_proxy_candidates_ranked.csv.

Raw DepMap files remain local and are not committed to GitHub.

Run order

```
python melanoma_depmap_v2.py  
python sox10_proxy_targets.py  
python sox10_proxy_plot_clean.py
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One-sentence conclusion

SOX10 was the strongest melanoma-selective dependency, and co-dependency plus tractability filtering nominated AMD1, PGM3 and LCMT1 as candidate druggable entry points into the SOX10-dependent melanoma state.